

Lab 2: SQL JOIN Lab

Overview

This lab will help you practice various SQL JOIN operations including INNER JOIN, LEFT JOIN, RIGHT JOIN, FULL OUTER JOIN, CROSS JOIN, and SELF JOIN.

You will use provided CSV files (`employees.csv` and `departments.csv`) to populate tables in your SQL database. After completing the exercises, submit your SQL queries to the classroom portal.

CSV Dataset Schema

employees.csv

- `employee_id` (integer)
- `first_name` (string)
- `last_name` (string)
- `department_id` (integer, nullable)
- `manager_id` (integer, nullable)
- `salary` (integer)
- `hire_date` (date)

departments.csv

- `department_id` (integer)
- `department_name` (string)
- `location` (string)
- `budget` (integer)

Exercises

1. INNER JOIN

- Select employee names, salaries, and their respective department names. Only include employees assigned to a department.

2. LEFT JOIN

- List all employees along with their department names. Show 'Unassigned' for employees without a department.

3. RIGHT JOIN

- Display each department name along with employee names. Include departments that have no employees.

4. FULL OUTER JOIN

- Select all employees and all departments, including employees without departments and departments without employees.

5. CROSS JOIN

- Create a query to show all possible combinations of employee names and department names.

6. SELF JOIN

- Using a self join, write a query to display each employee along with their manager's name. For employees without a manager, display 'No Manager'.

7. INNER JOIN with Condition

- Select employees whose salary is above the average salary of their department.

8. LEFT JOIN with Aggregation

- Calculate the average salary for each department. Include departments with no employees, showing a count of 0 and an average salary as 'NULL'.

9. RIGHT JOIN with Aggregation

- Count the number of employees in each department, including departments with no employees. Sort by the number of employees in descending order.

10. FULL OUTER JOIN with Advanced Conditions

- Write a query to select all employees and department names, including employees without departments and departments without employees. Display a meaningful placeholder for missing data.

Submission

Please complete each exercise by writing the corresponding SQL query. Submit your SQL queries as a single document. Upload your completed document to the classroom.